

Data Science for Business

L1: Introduction

Prof. Dr. Jella Pfeiffer



***Who are you? And what do you
know about Data Science & ML?***

Learning Objectives



- Understand the typical process of a data science project in business
- Know some typical data science methods
- Rough ideas about how this knowledge can be used in the business context
- ...and all this with hands on experience in Python



Presentation of Research Group – Intelligent & Responsible Information Systems (IRIS)



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Five Research Topics

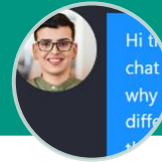
- Determining an appropriate point of time when to offer help
- Predict the Context and User Preferences
- User interaction with fully digital LLM-based Avatars

Intelligent User Assistance in Virtual Realities



- How human-like should CAs be designed?
- When do users trust CAs and why?
- Can we incorporate LLM-based CAs in Experimental Research?

Conversational Agents



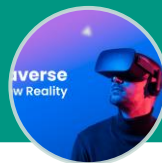
- How do users perceive fairness of machine learning algorithms?
- What effect does the EU AI Act have on companies?

Fairness In AI



- In which industries is XR applied?
- Balancing User Privacy and Business Benefits

Consequences and Design of the Metaverse



- How should an XR-based public participation app be designed to enable citizens' participation
- What challenges and practical solutions exist regarding data sharing under the CSRD?
- What value can be achieved beyond mere reporting?

E-Participation and Sustainability

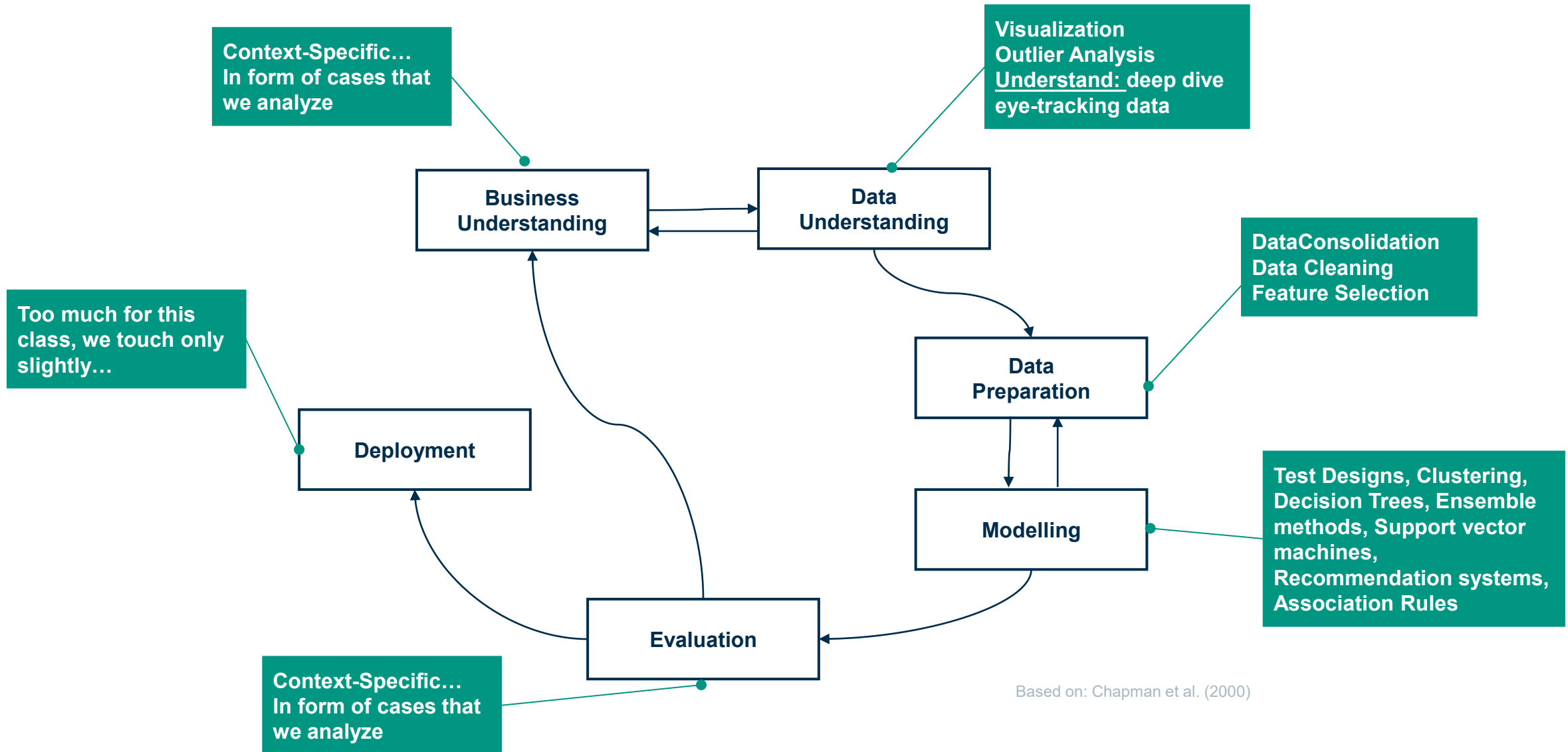


Where to find us:



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Content and Agenda of Class



Schedule

Date	8:00-9:30 bzw Slot 1	9:45-11:15 bzw Slot 2	11:30-13:00 bzw Slot 3
22.04.2026		L1: Organization, Data Science Definitions, Case Introduction	* Extra Exercise (optional): Only for Beginners: Gentle Introduction into Python
29.04.2026	L2: CRISP, Business Understanding and Data Understanding		Exercise 1: Handling and Understanding Data with Pandas, Group Formation Case
06.05.2026	L3: Research Deep Dive Eye-Tracking Data, Data Preparation and Feature Selection	L4: Test Designs for Modelling : Training, Testing	Exercise 2: Visualize Data & Data Understanding of Eye-Tracking-Data
13.05.2026	L5: Modelling Methods 1 : Tuning, Decision Trees	L6: Modelling Methods 2 : Random Forest (& Support Vector Machines)	Exercise 3: Decision Trees & Random Forests
20.05.2026	L7: Modelling Methods 3 : Recommender Systems	L8: Modelling Methods 4 : Unsupervised Learning	Exercise 4: Recommender Systems
03.06.2026			
10.06.2026	L9: Evaluation with in Class-Case: Churn Prediction	L10: Research Deep Dive: Data Science for Virtual Reality Eye-Tracking Data	Exercise 5: Unsupervised Learning + Eye Tracking
17.06.2026		Q&A Case (optional)	
24.06.2026			
01.07.2026		Q&A Case (optional)	
08.07.2026			
15.07.2026		Q&A Case (optional)	
22.07.2026 & 23.07.2026	Case Presentation (23.07.2026: Institute for Information Systems 5C-01)		
29.07.2026		L: Q&A Exam	

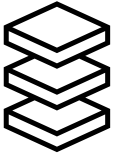
Case Introduction

Goal?



- Applying the learned skills to a real-world dataset
- Learning new skills
- Build a business case around the dataset

What?



- Analyze an Airbnb dataset with listings, reviews, and pricing calendars across cities
- 12 cities of Inside Airbnb
- Group Formation will be done in the Exercise on 29.04.2026

Grading?



- Presentation and code book will be graded
- The Case Study makes up 40% of the final grade

Case Introduction

General Task

- Data Preparation: How do you clean and preprocess the data?
- Feature Extraction: What insights can you derive from the dataset? Which are important for your goal?
- Feature Selection: What features are actual important/usefull?
- Method Selection: Which approach best fits your goal?
- Reasoning: Why did you choose a specific method?

Focus Area

- Each group has to select one focus area and one city

Case Introduction: Focus Area

Price Prediction



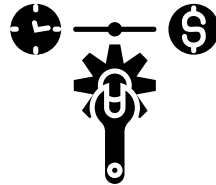
- Predict the price of a listing to a certain date
- Compare multiple methods

Classification



- Predict binary and multi classes
- Compare multiple methods

Model Evaluation



- Explore different test & train splits
- Explore different cross-validation methods
- Compare the approaches

Content-Based Recommender

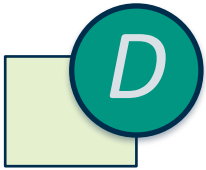


- Solve the cold start issue
- Calculate similarities
- Make recommendations

Exam

- 10.08.2026, 12:30-13:00, 30.22 Wolfgang-Gaede-Hörsaal
- Content: Mixture of concepts learned in lecture and python-tasks (correcting code, finding mistakes, interpreting outputs and mini-programming tasks)
- Q&A in last lecture

Symbols on Class Slides



Definition



Goals of this lecture



References to other lectures and exercises



Summary at the end of each lecture



Topic of next lecture

Source: Carr (2003)

Sources used for this slide



Quiz, discussion, ...



Quote



Important hint

Master Classes at IRIS

Data Science for Business (English)

- Every summer term
- Focus on Consumer and User Data
- Understanding of Consumer Data (also eye-tracking data)
- CRISP-Cycle
- Methods:
 - Unsupervised: Clustering, Association Rules
 - Supervised: Classification (Random Forests and Variants, optional: Support Vector Machines), Recommender Systems
- All **Applied with Python**
- Larger Case Study with Python (40% of grade)

Responsible AI (deutsch)

- Every winter term
- Focus on trustworthiness and ethical as well as moral considerations of AI
- AI Regulation (EU AI Act) and others
- Privacy & Security Aspects
- Highly discussion-focussed class:
 - Preparation with papers and other material
 - Discussions in class

What this class addresses and what it doesn't

Industry sector	11: Agriculture, forestry and fishing	21: Mining, Quarrying, and Oil & Gas	22: Utilities	23: Construction	31-33: Manufacturing	42: Wholesale Trade	44-45: Retail Trade	48-49: Transportation and Warehousing	51: Information	52: Finance and Insurance
	53: Real Estate and Leasing	54: Professional, Scientific and Techn. Services	55: Management of Companies and Enterprises	56: Ad. Support & Waste Mgmt. & Remediation	61: Educational Services	62: Health Care and Social Assistance	71: Arts, Entertainment, and Recreation	72: Accommodation & Food Services	81: Other Service (except Public Admin.)	92: Public Administration
Functional area	Production-operations	R&D & Engineering	Accounting & Finance	Management & Admin.	Marketing & Sales	Law	Personnel & Labor Relat.	IT		
AI category	Descriptive		Predictive		Prescriptive		Generative			
Input data	Numerical		Text		Audio and Signal		Images			
Learning behavior	Supervised learning		Unsupervised learning		Semi-supervised learning		Reinforcement learning			
Deep learning	Yes					No				
Explainable AI	Yes					No				
AI algorithm	Bayesian learning	Logistic regression	k-nearest neighbors	SVM	DT and RF	Boosting	Autoencoder	k-means		
	LDA	MLP	CNN	RNN	GAN	Reinforcem. learning	Language models	Other algorithm		

At WIN:

Responsible AI

Artificial Intelligence in Service Systems



Artificial Intelligence in Service Systems - Generative AI Applications and Adoption

(Gen)AI-based Automation in Organizations

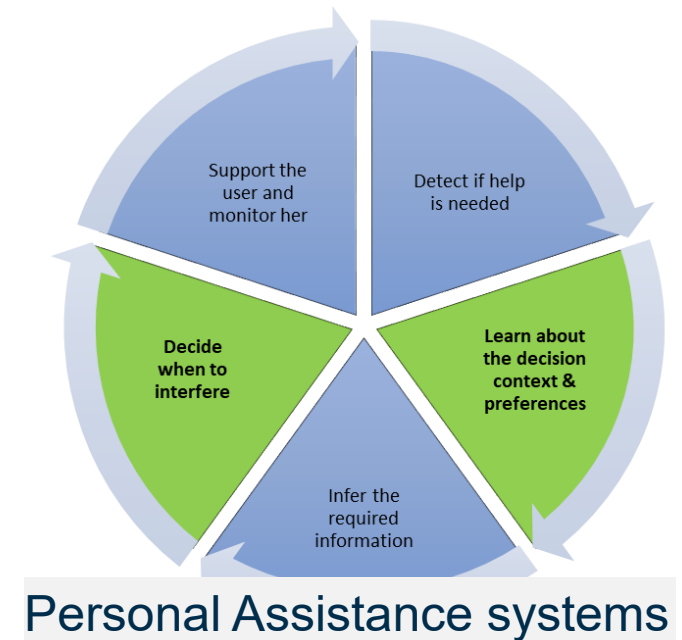
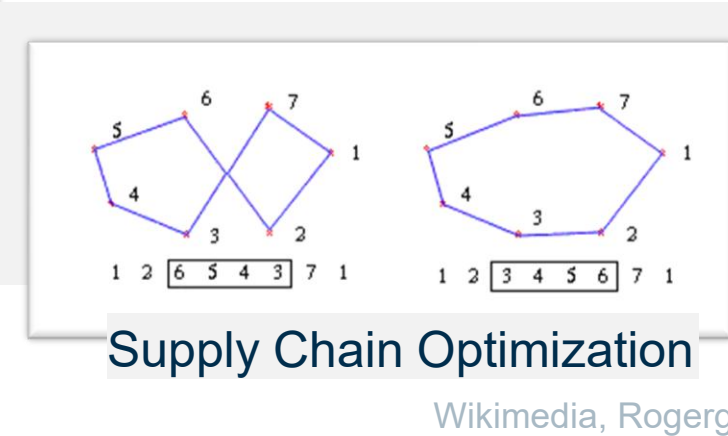
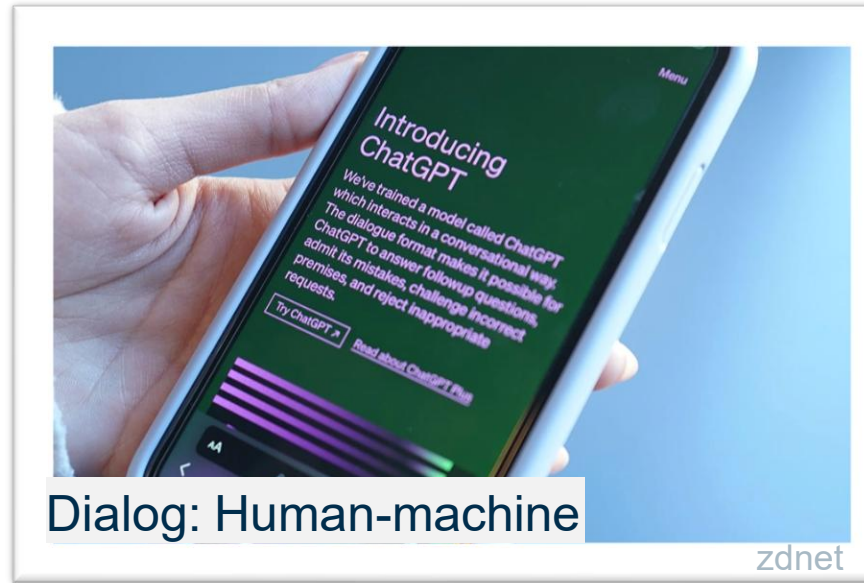
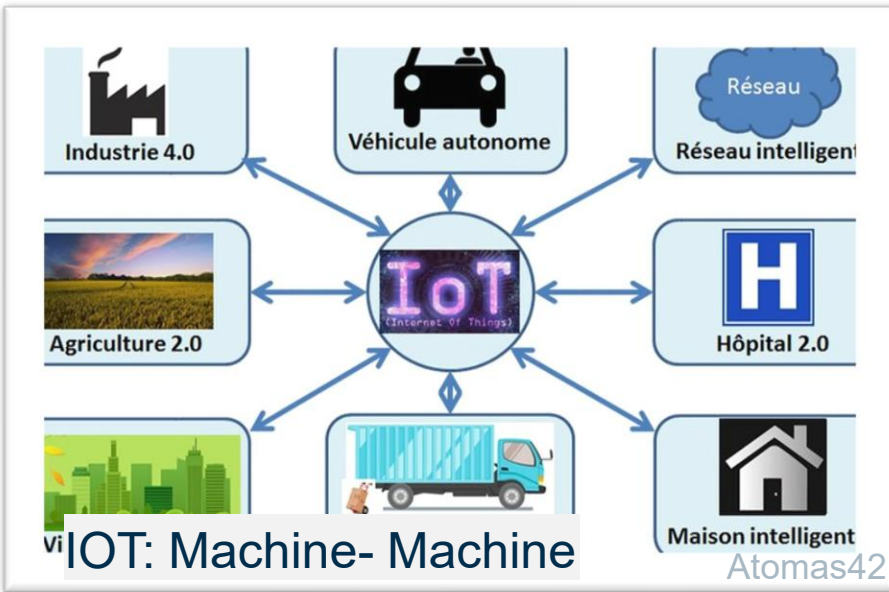
Bendig & Bräunche (2025)

What it doesn't (because no GenAI)

	Model level	System level	Application level
Output Modality (Selection)	<i>Underlying AI model for different data modalities (e.g., image, text, code)</i>	<i>Embedding model functionality to provide interface for interaction</i>	<i>Solving dedicated business problems and stakeholder needs</i>
Text Generation	X-to-text models, e.g., GPT-4 and LLaMA 2	Conversational agents and search engines, e.g., ChatGPT and YouChat	- Content generation (e.g., SEO and customer service) - Translation and text summarization
Image/Video Generation	X-to-image models, e.g., Stable Diffusion and DALL-E 2	Image/video generation systems and bots, e.g., Runway and Midjourney	- Synthetic product and advertising visuals - Educational content
Speech/Music Generation	X-to-music/speech models, e.g., MusicLM and VALL-E	Speech generation systems, e.g., ElevenLabs	- AI music generation - Text-to-speech generation (e.g., news, product tutorials, etc.)
Code Generation	X-to-code models, e.g., Codex and AlphaCode	Programming code generation systems, e.g., GitHub Copilot	- Software development - Code synthesis, review, and documentation

Feuerriegel et al. (2024)

What we do: Data Science Applications in Business



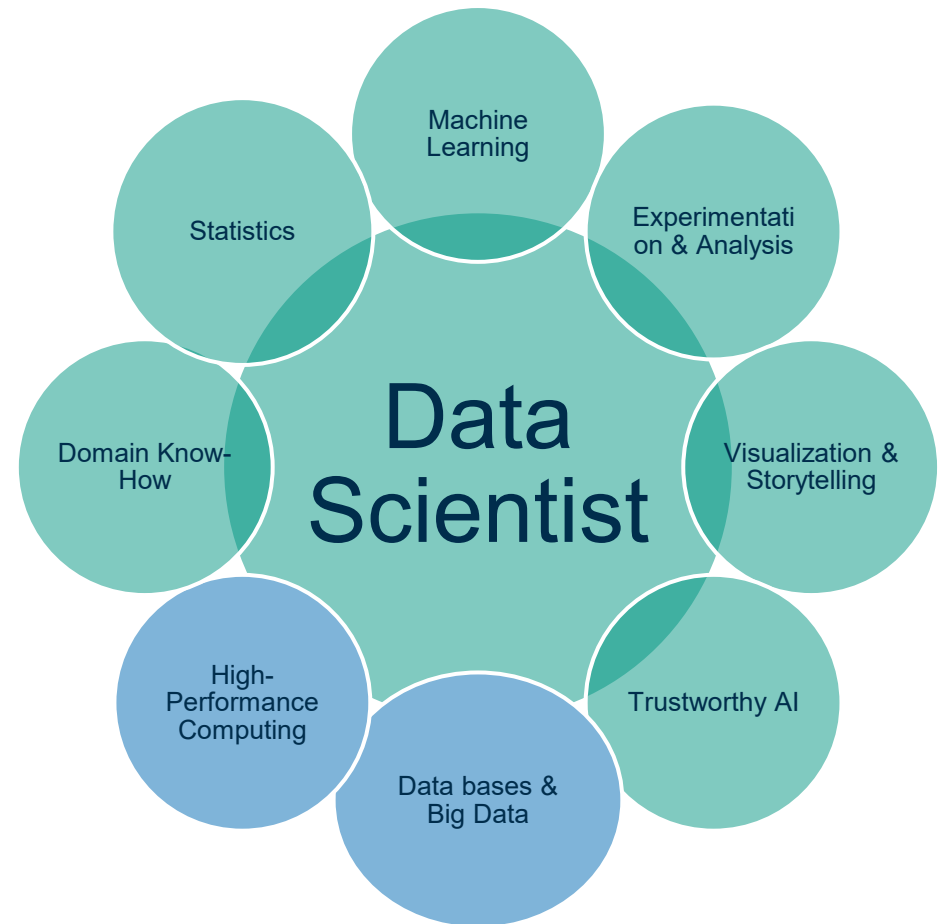
<https://identityweek.net/visa-announces-generative-ai-powered-fraud-solution-to-combat-account-attacks/>

<https://www.heute.at/i/so-spart-man-beim-handyvertrag-bis-zu-80-prozent-120027616/doc-1hp3gqir1>

Data Science: Definition and Necessary Capabilities

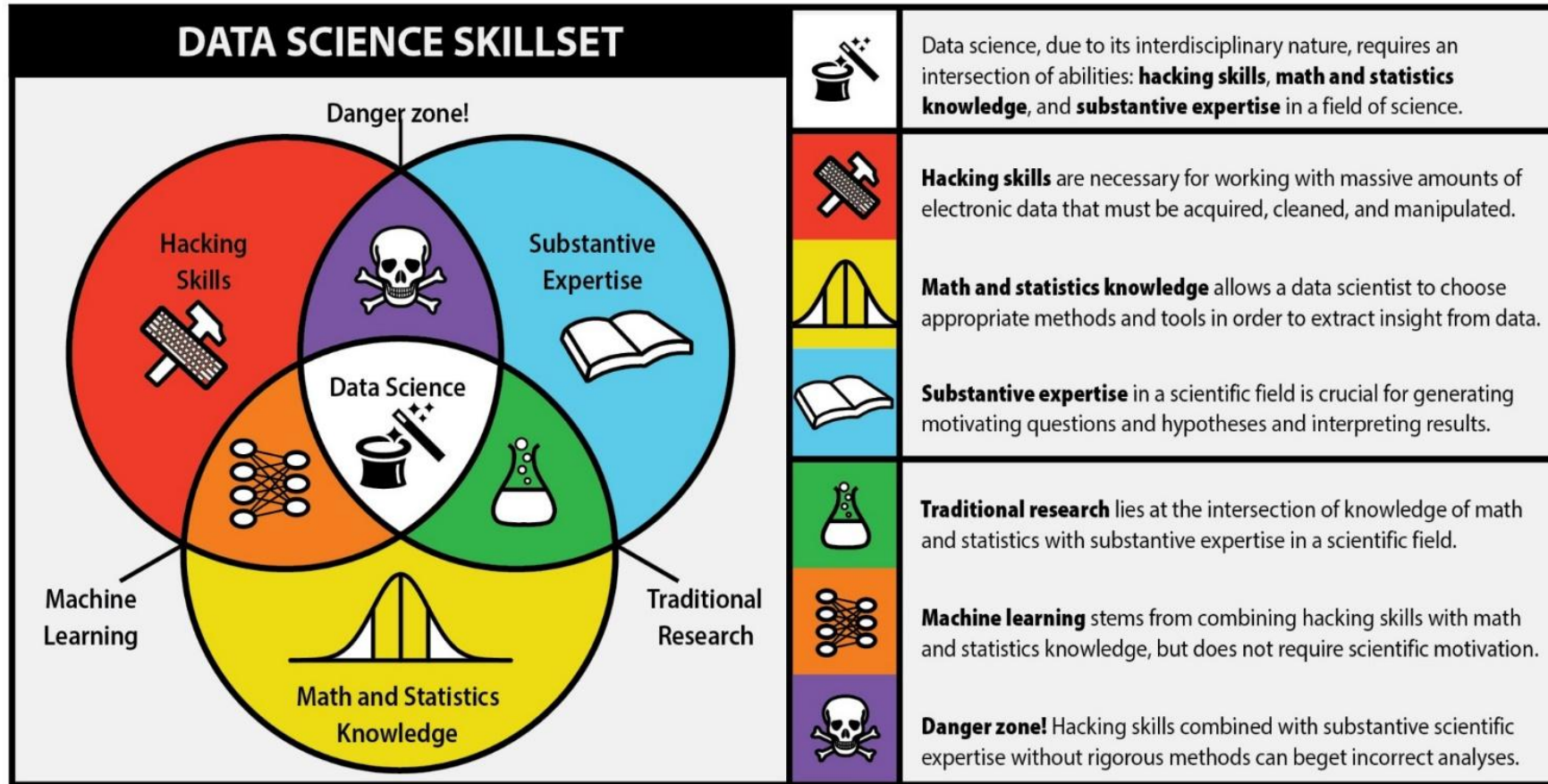
D

Data science means the scientific study of the creation, validation and transformation of data to create meaning.[...] Data science uses scientific principles to get meaning from data and uses machine learning and algorithms to manage and extract actionable, valuable intelligence from large data sets. [definition from the data science association]



Own view adapted from Haneke et al. (2021)
und Sharda et al. (2021)

Data science skillset and danger zone



Analytics Or Data Science?

- Some see differences, some don't
- One view:
 - **Data analyst** is just another term for professionals who were doing Business Intelligence in the form of data compilation, cleaning, reporting, and perhaps some visualization. Skill sets: Excel, SQL, Reporting → More descriptive; but some also deeper knowledge in statistical modeling, experimentation, and analysis.
 - **Data scientist**: responsible for predictive analysis, statistical analysis, and use of more advanced analytical tools and algorithms. They may have a deeper knowledge of algorithms and may recognize them under various labels—data mining, knowledge discovery, or machine learning. Also, potentially knowledge on optimization (operations research methods).

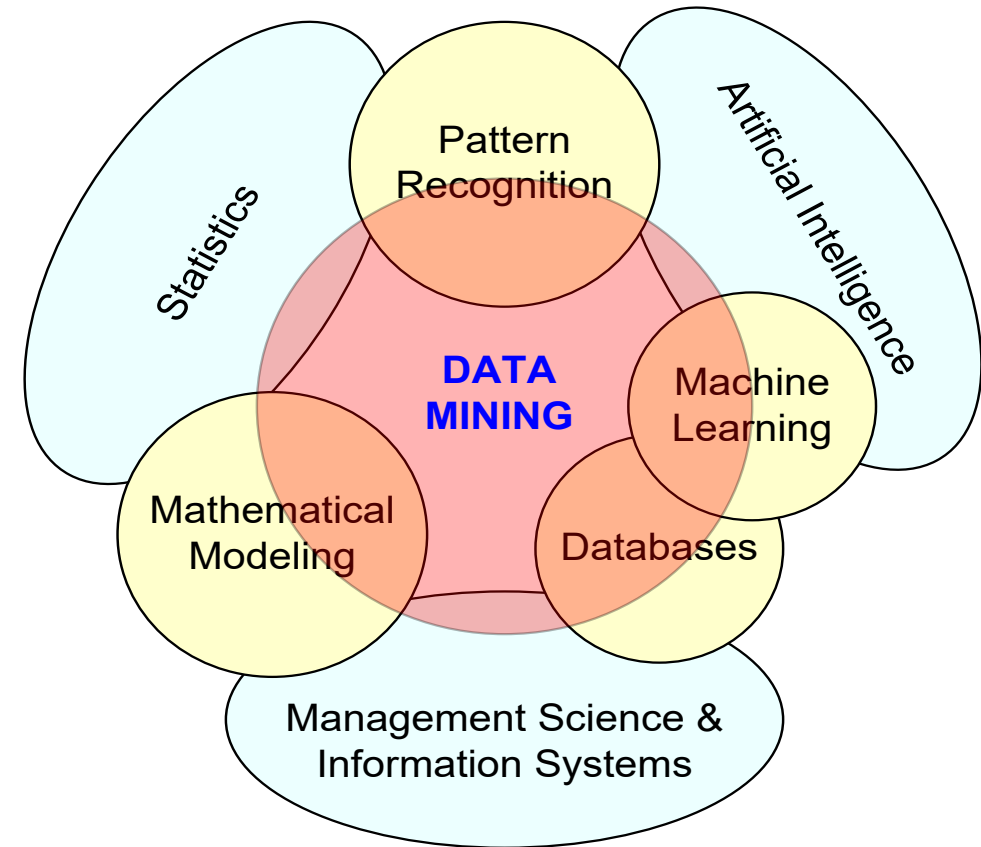
..and what's a data engineer?

Data Scientist also known as Data Managers, statisticians. 	Data Engineers also known as database administrators and data architects. 	Data Analysts also known as business Analysts. 
A data scientist will be able to take data science projects from end to end. They can help store large amounts of data, create predictive modelling processes and present the findings.	They are versatile generalists who use computer science to help process large datasets. They typically focus on coding, cleaning up data sets, and implementing requests that come from data scientists.	They typically help people from across the company understand specific queries with charts.
Skills: Mathematics, Programming, Communication 	Skills: Programming, Mathematics, Big data 	Skills: Statistics, Communication, Business knowledge 
<i>Will use programmes such as:</i> SQL, Python, R	<i>Will use programmes such as:</i> Hadoop, NoSQL, and Python	<i>Will use programmes such as:</i> Excel, Tableau, SQL

Data Mining Or Data Science?

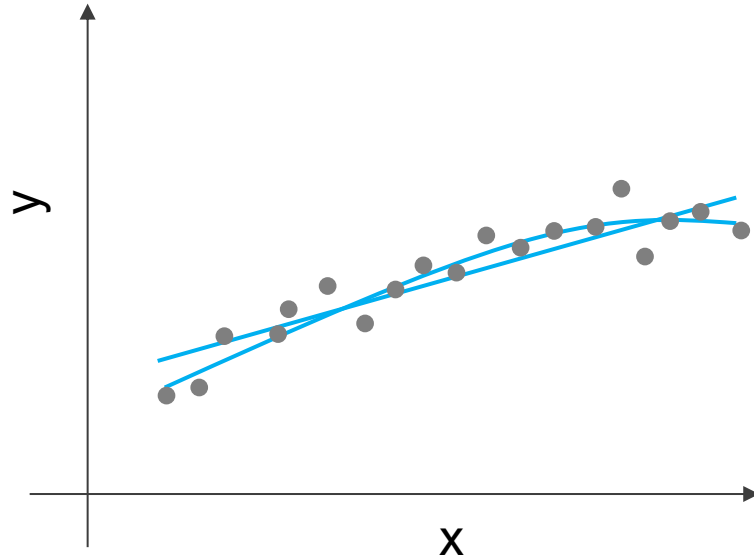
Data Mining: The nontrivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data stored in structured databases.

Fayyad et al., (1996)



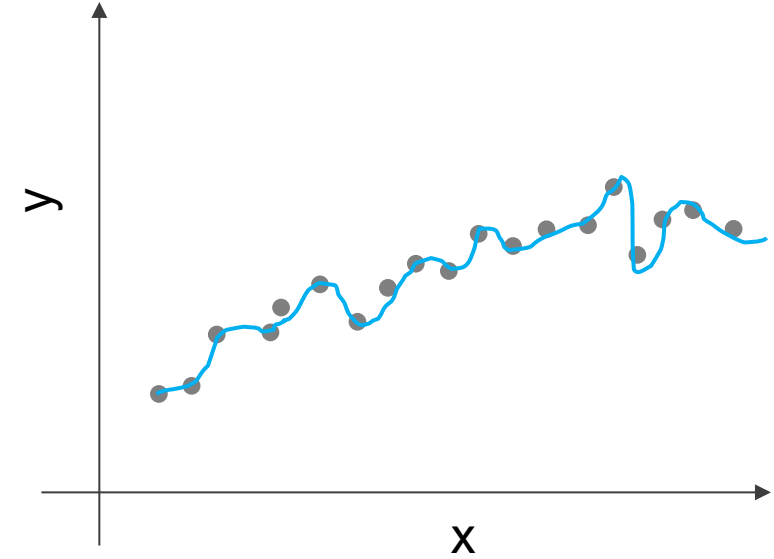
Source: Sharda et al. 2014, Figure 5.1

Predicting vs. Explaining



Explaining

- General **influence** of one on another variable ($y \sim x$)
- Don't model too complex relations
- Usually simple models (linear, quadratic, 2-way interaction)
- Use all data to fit model
- Accuracy is secondary (R^2)
- Anchored in theory



Predicting

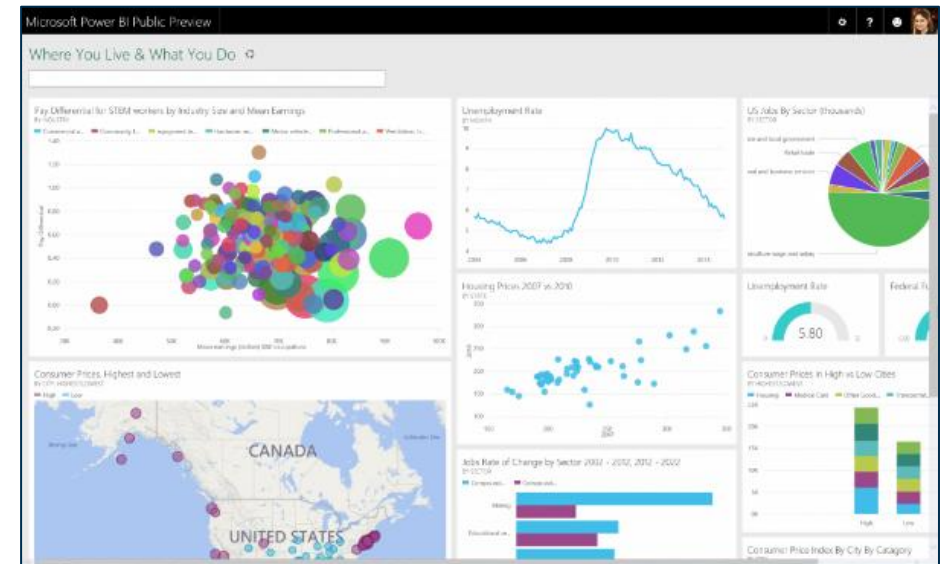
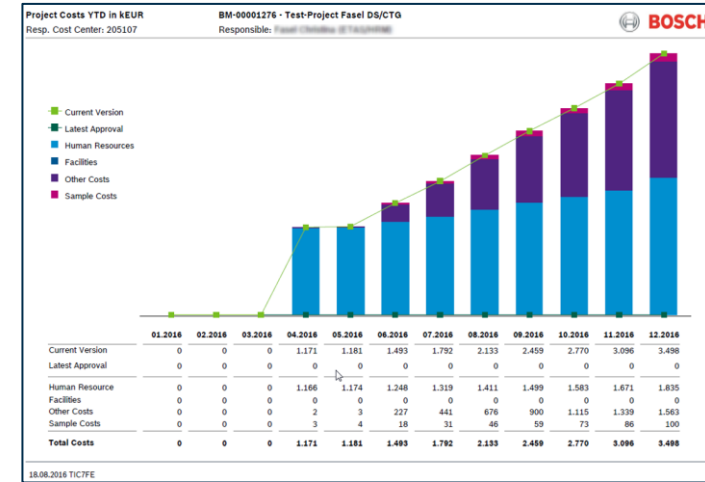
- Focus on target variable: predict y (at any cost)
- Any relation is possible
- Highly complex models (black box)
- Hold back data to evaluate fit (to avoid overfitting)
- Accuracy first
- Focus on practical application

Business Intelligence & Analytics or Data Science?

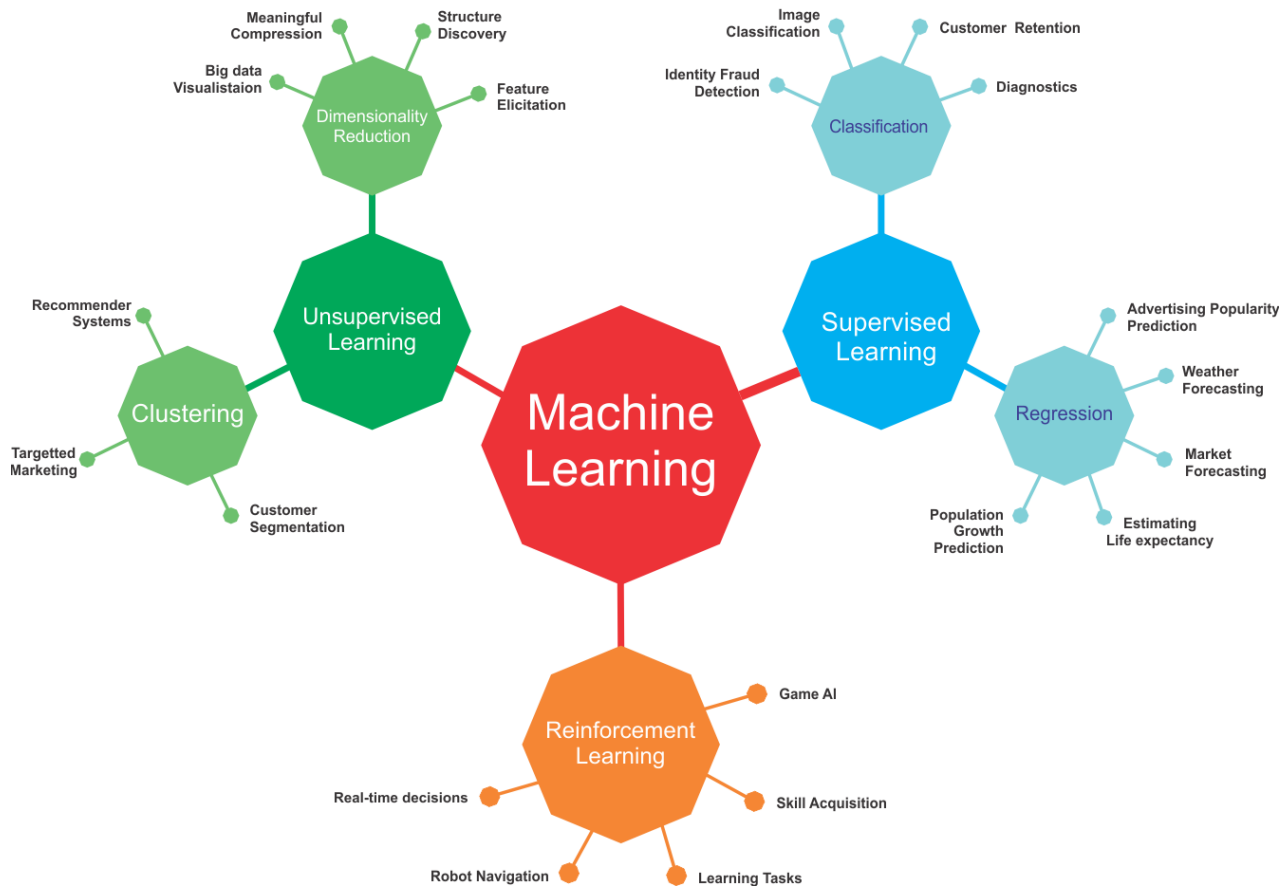
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Definition: Business Intelligence and Analytics (BI&A)

BI&A refers to the techniques, technologies, systems, practices, methodologies, and applications that analyze critical business data to help an enterprise better understand its business and market and make timely business decisions.



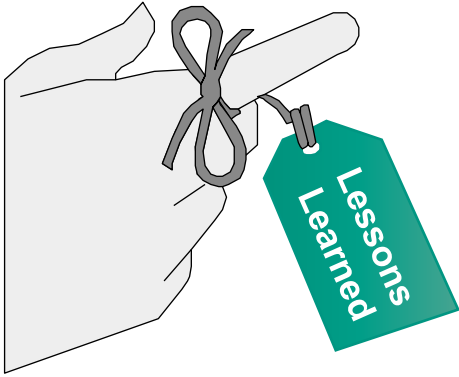
This Course focusses on machine learning



Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed
Arthur Samuel, 1959

A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .
Tom Mitchell, 1997

Summary



- CRISP is one process model one can follow when doing a data mining project
- We follow this model throughout the lecture and you learn how to successfully conduct a data mining/data science project
- The first steps define the success of the later steps and use a lot of the time and money
- The test design and the right choice of accuracy measurement of your model is very important for success



- CRISP, Business Understanding and Data Understanding

Literature

- Bendig, D., & Bräunche, A. (2025). The role of artificial intelligence algorithms in information systems research: a conceptual overview and avenues for research. *Management review quarterly*, 75(4), 2863-2908.
- Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative AI: S. Feuerriegel et al. *Business & Information Systems Engineering*, 66(1), 111-126.
- Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). CRISP-DM 1.0 Step-by-step data mining guide.
- Chen, H., Chiang, R. H. L, and Storey, V. C. 2012. “Business Intelligence and Analytics: From Big Data to Big Impact,” *MIS Quarterly* (36:4), pp. 1165-1188.
- Sharda, R., Delen, D., Turban, E., Aronson, J., & Liang, T. P. (2014). *Business Intelligence and Analytics: Systems for Decision Support*. London: Prentice Hall.
- Sharda, R., Delen, D., & Turban, E. (2021). *Analytics, Data Science, & Artificial Intelligence: Systems for Decision Support*. Harlow: Pearson.